

## jatin singh shekhawat

BK Birla Institute of Engineering Technology

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### ACADEMIC DETAILS

- **BK BIRLA INSTITUTE OF ENGINEERING AND TECHNOLOGY, PILANI** [2022-26]
  - B Tech ECE (8.43 CGPA)
- **STAR ACADEMY, JHUNJHUNU** [2021-22]
  - Senior Secondary (RBSE)
- **STAR ACADEMY, JHUNJHUNU** [2019-20]
  - Matriculation (RBSE)

### FIELDS OF INTEREST

- Embedded Systems, IoT, and Hardware-Software Co-Design for automation, wireless control, and smart device communication.

### TECHNICAL SKILLS

- **Programming language** (C, Python)
- **Development Tools IDEs** (Arduino IDE, ModelSim, Xilinx, TCLsh, EDA Playground, Visual Studio Code)
- **Software for Projects** (Inkscape, Universal Gcode Sender, Drawbot)
- **Communication Protocols** (UART, Wi-Fi (ESP8266), IR Remote, Bluetooth (HC-05))
- **Areas of Expertise** (Embedded Systems, IoT Integration, G-code Automation, Real-time Control Systems)

### CERTIFICATES

- Successfully completed a 45-day training program on "Advanced Digital Design using HDL." Gained practical experience in digital circuit design and received a certification for the course
- Successfully achieved proficiency in the Introduction to the Internet of Things and Embedded Systems through an online course on Coursera.
  - [view certificate](#)

### MAJOR PROJECTS AND SEMINAR

- **Voice-Activated CNC Machine**  
(Voice-Controlled IoT System, (Q1 2025))
  - Built handwriting CNC machine using Arduino, ESP8266, stepper motors, and 3D-printed parts.
  - Developed Arduino firmware to interpret G-code and control writing.
  - Successfully connected CNC via HC-05 Bluetooth before moving to ESP8266 Wi-Fi.
  - Integrated Google Assistant and IFTTT for voice-triggered CNC operation.
  - Achieved fully wireless, voice-controlled handwriting automation.
- **IR-Based Wireless Transmitter**  
(IoT Embedded Project, (Jun'2025))
  - Built an IR remote emulator using Arduino UNO and IRremote library to capture and transmit signals.
  - Extended the project using ESP8266 to host a web interface for wireless IR control via browser.
  - Used IRremoteESP8266 to decode and send IR codes directly from ESP8266, eliminating the need for Arduino.
  - Enabled smart control of IR-based appliances over Wi-Fi using mobile or PC.

- **Reflow Oven Temperature Controller**

*(Thermal Automation System, (Q3 2025))*

- Designed a multi-stage reflow oven controller using Arduino, NTC thermistor, and PID logic.
- Automated Preheat, Soak, Reflow, and Cooling with real-time CSV logging via Serial.
- Implemented transition summaries (e.g., "Reached 45.0°C from 30.0°C in 52s") for thermal analysis.
- Adapted timings dynamically to match ambient temperature conditions.

## TRAINING AND INTERNSHIPS

- **In-House Summer Internship - BK Birla Institute of Engineering Technology (12th-24th June 2023)**

- Successfully completed a summer internship organized by the Department of Applied Sciences Humanities.
- Gained hands-on experience in relevant engineering concepts and practical applications
- [view certificate](#)

- **Advanced Digital Design using HDL - FutureWiz (7th June 2024 - 2nd August 2024)**

- Completed 45-day training focused on HDL-based digital design.
- Gained hands-on experience in synthesis, simulation, and verification techniques.
- [view certificate](#)

## STRENGTHS

- Social Interaction, Hardworking.

## INTEREST AND HOBBIES

- Solving Puzzles.
- Playing Badminton